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GM INSTITUTE OF TECHNOLOGY, DAVANGERE

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DEPARTMENT OF ENGINEERING PHYSICS

Module- 5

Applications of Physics in Animation & Computing

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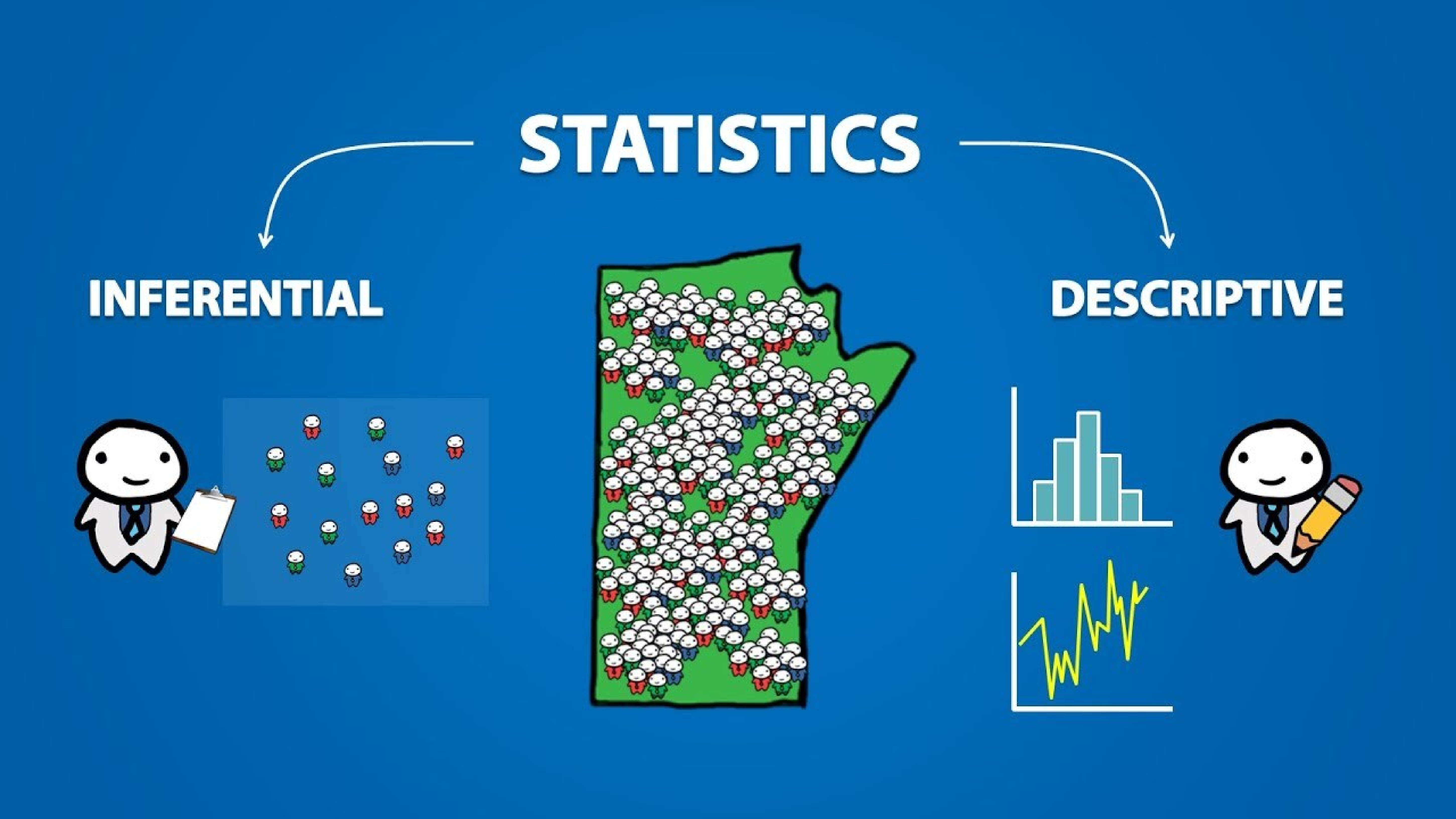
Chapter-2

Stattistical Physics for Computing

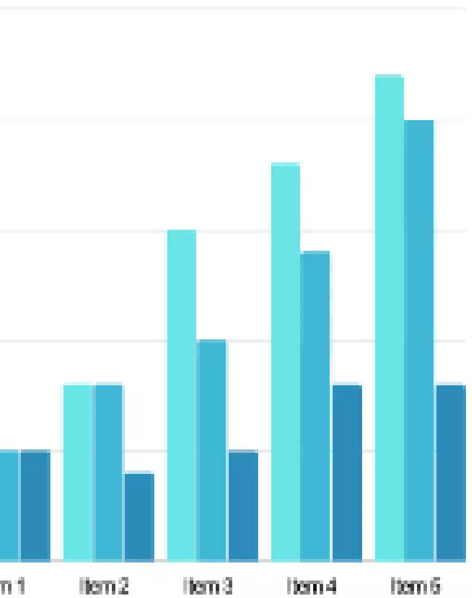
Syllabus

Physics of Animation: Taxonomy of physics based animation methods, Frames, Frames per Second, Size and Scale, Weight and Strength, Motion and Timing in Animations, Constant Force and Acceleration, The Odd rule, Odd rule Scenarios, Motion Graphs, Examples of Character Animation : Jumping, Parts of Jump, Jump Magnification, Stop Time, Walking: Strides and Steps, Walk Timing. Numerical Problems

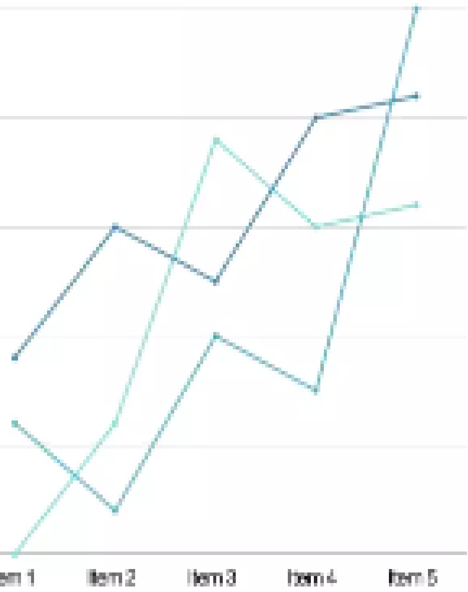
Statistical Physics for Computing : Descriptive statistics and inferential statistics, Poisson distribution and modeling the probability of proton decay, Normal Distributions (Bell Curves), Monte Carlo Method : Determination of Value of π. **Numerical Problems.**


STATISTICS
INFERENTIAL
DESCRIPTIVE

Introduction



Descriptive statistics are numbers that are used to summarize and describe data. The word "data" refers to the information that has been collected from an experiment, a survey, a historical record, etc.

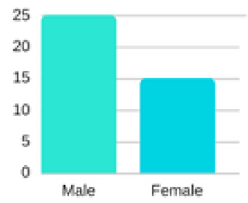


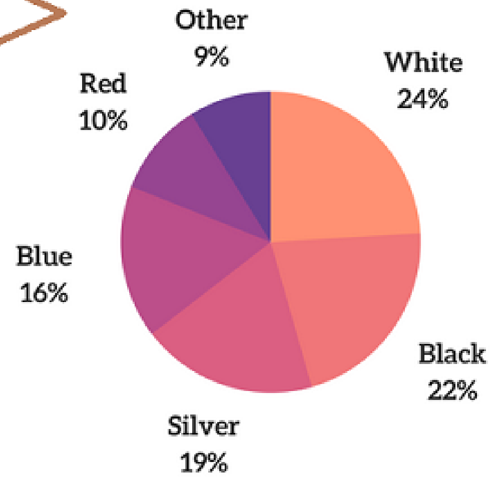
Inferential statistics is a set of data taken from the population to represent the population. Inferential statistics uses a random sample of data taken from a population to describe and make inferences about the population.

Example

	A	B	C	D
1	Respondent Number	Age	Gender	Favorite Car Color
2	1	22	M	White
3	2	37	F	Silver
4	3	45	F	Black
5	4	62	F	Gray
6	5	28	M	Red
7	6	45	M	Green
8	7	88	F	Brown
9	8	61	M	White
10	9	95	M	Black
11	10	27	M	White
12	11	39	F	Green
13	12	43	M	Brown
14	13	55	F	Black
15	14	59	F	White

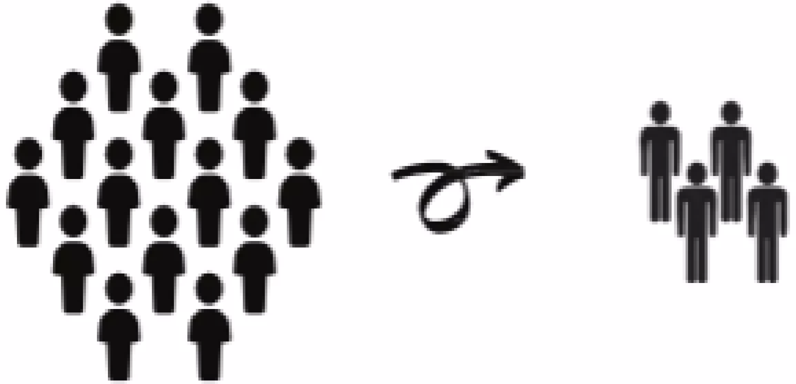
RAW DATA





Descriptive Statistics

Inferential statistics

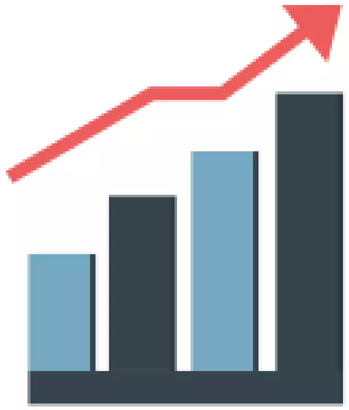


Descriptive and Inferential statistics

Q: Distinguish between descriptive and inferential statistics (**6M**)

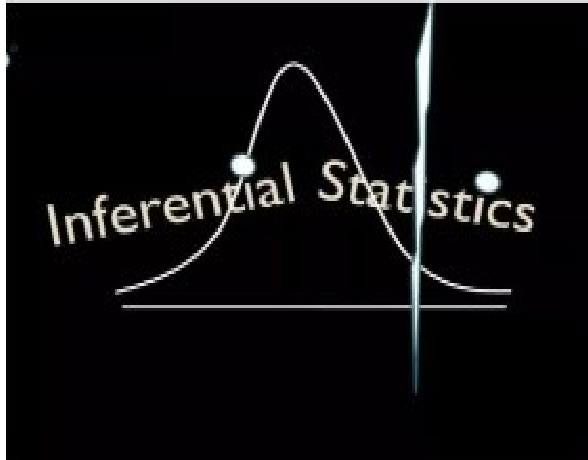
Descriptive Statistics

- It gives information that describes the data in some detailed manner.
- It helps in organizing, analyzing and to present data in a meaningful manner.
- It is used to describe a situation.
- It explains already known data and is limited to a sample or population having a small size.
- It can be achieved with the help of **charts, graphs, tables**, etc.



Inferential Statistics

- It makes inferences about the data drawn from exp, history or survey, etc
- It allows us to compare data and make hypotheses and predictions
- It is used to explain the chance of occurrence of an event.
- It attempts to reach the conclusion about the population.
- It can be achieved by **probability**.



Descriptive and Inferential statistics

Descriptive Statistics	Inferential Statistics
Works with a smaller data set	Works with a large data set
Process is simpler to do	Process in more complex as we have to decide on the best sampling techniques
Results obtained represent the entire data set	Results obtained represent a portion of the population, but can be used to deduce information about the entire population with some amount of uncertainty
Error involved is usually less	Error involved is usually more

Statistics: Mean / Median /Mode/ Variance /Standard Deviation

Example

Let's consider heights of persons.

Heights = [168,170,150,160,182,140,175,191,152,150]

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[140, 150, 150, 152, 160, 168, 170, 175, 182, 191]
```

Mean	Median	Mode	Variance	Standard Deviation
Mean is also known as average of all the numbers in the data set Ans: **163.8**	Median is mid value in the ordered data set. Ans: **168**	Mode is the number which occur most often in the data set. Ans: **150**	VARIANCE measures how far the values of the data set are from the mean. The average of the squared deviations is the population variance and denoted by sigma-squared(σ^2)	It is square root of the variance and denoted by Sigma (σ)

Poisson Distribution

- Poisson distribution is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space, occur with a known constant mean rate.

- It is named after French mathematician Siméon Denis Poisson-1837

- **For instance**, a call center receives an average of 180 calls per hour, 24 hours a day. The calls are independent; receiving one does not change the probability of when the next one will arrive. The number of calls received during any minute has a Poisson probability distribution with **mean 3**: the most likely numbers are **2 and 3** but **1 and 4** are also likely and there is a small probability of it being as low as **zero** and a very small probability it could be **10**.

- Another example is the number of decay events that occur from a radioactive source during a defined observation period.

Poisson Distribution

"If the probability **P** is so **small** that the function has significant value only for very small ***k***, then the distribution of events can be approximated by the ***Poisson Distribution***".

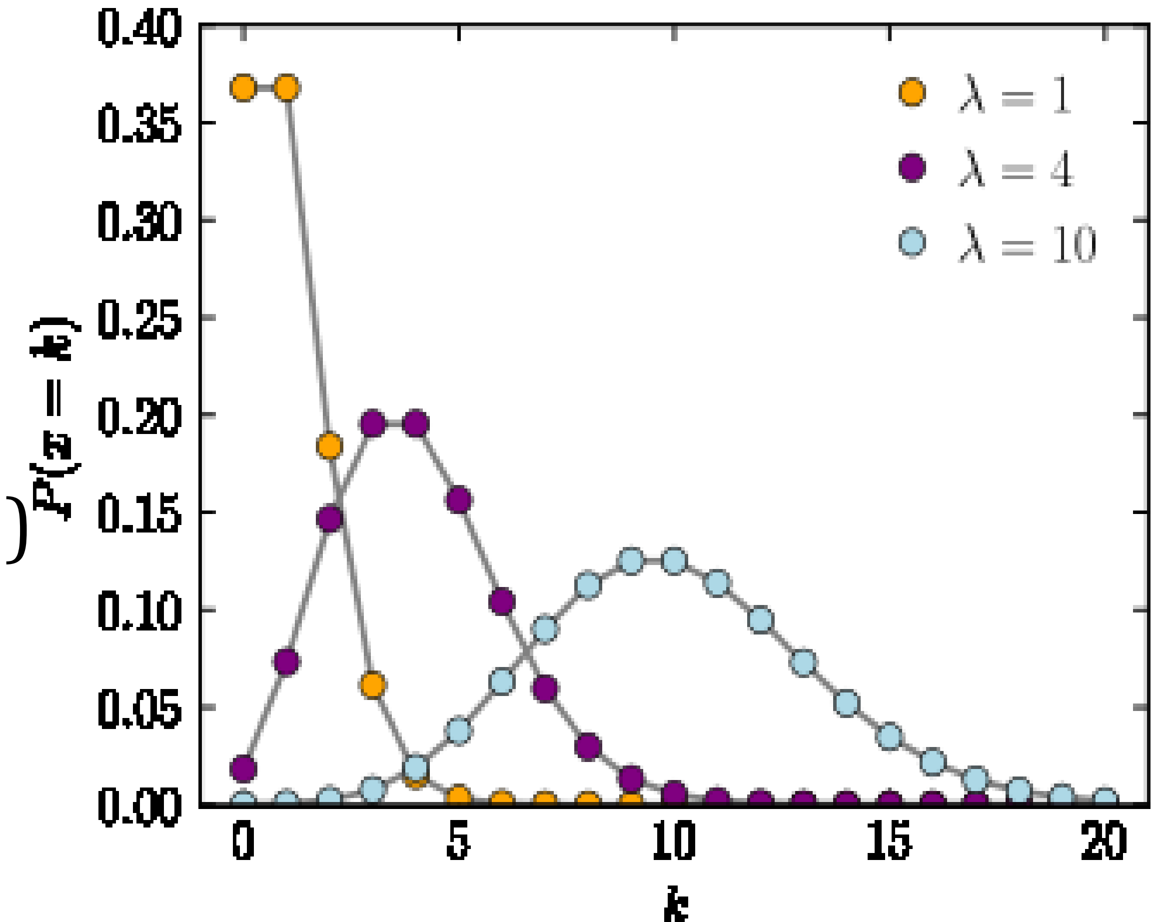
Probability function (P):

A discrete Radom variable ***X*** is said to have a Poisson distribution, with parameter $\boldsymbol{\lambda} > 0$, if it has a probability function as;

$$P_{(X=k)} = \frac{\boldsymbol{\lambda}^k \cdot e^{-\boldsymbol{\lambda}}}{k!}$$

where

- ***k*** is the number of occurrences (***k*** = 0, 1,2,3,4,......)
- $\boldsymbol{\lambda}$ is equal to the expected value of **X** and also to its **Variance**



Example: probability for Poisson distributions

On a particular river, overflow floods occur once every 100 years on average. Calculate the probability of k = 0, 1, 2, 3, 4, 5, or 6 overflow floods in a 100 year interval, assuming the Poisson model is appropriate.

Solution: Because the average event rate is one overflow flood per 100 years, $\boldsymbol{\lambda = 1}$.

$$P(\text{k overflow floods in 100 years}) = \frac{\lambda^k e^{-\lambda}}{k!} = \frac{1^k e^{-1}}{k!}$$

$$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$P(\text{k=0 overflow floods in 100 years}) = \frac{\lambda^k e^{-\lambda}}{k!} = \frac{1^0 e^{-1}}{0!} = \frac{e^{-1}}{1} = 0.368$$

$$P(\text{k=1 overflow floods in 100 years}) = \frac{\lambda^k e^{-\lambda}}{k!} = \frac{1^1 e^{-1}}{1!} = \frac{e^{-1}}{1} = 0.368$$

$$P(\text{k=2 overflow floods in 100 years}) = \frac{\lambda^k e^{-\lambda}}{k!} = \frac{1^2 e^{-1}}{2!} = \frac{e^{-1}}{2} = 0.184$$

Poisson Distribution: Modeling the Probability for Proton Decay

- Probability for observing a proton decay can be estimated from **Poisson distribution**.
- If the lower bound lifetime is projected to be of the order of $\boldsymbol{\tau = 10^{33}}$ ***Yrs*** for Proton Decay,
- Then No. of protons $\boldsymbol{N}$ modeled by decay eqn;

$$N = N_0\, e^{-\lambda' t}$$

Here $\lambda' = 1/\tau = 10^{-33}/yr$ is the **decay constant** which gives proton decay probability in a year.

Since λ' is **so small**, then $e^{-\lambda t}$ can be represented ;

$$e^{-\lambda t} = 1 - \lambda' t \qquad \therefore\ N = N_0\,(1 - \lambda' t)$$

For small sample, observation of proton decay is too small, but suppose we consider the volume of protons represented by the Super Kameokande detector in Japan. The No. of protons in the detector volume is **7.5 x 10^{33}** protons (i.e., N_0).

For one year of observation, the No. of expected proton decays is then

$$N_0 - N = N_0\, \lambda' t = (7.5 \times 10^{33})(10^{-33}/yr)(1yr)$$

$$\therefore N_0 - N = 7.5$$

If **40%** area around the detector is covered by photodetectors, then we expect abt **3 observations** of proton decay/year based on a 10^{33}year lifetime.

So far, no convincing proton decay events have been seen. Poisson statistics conveniently provides this absence of observations.

If we presume that $\boldsymbol{\lambda = 3}$ **average** observed decays/yr, then from the **Poisson distribution** function, the probability for "**0**" observations of a decay is;

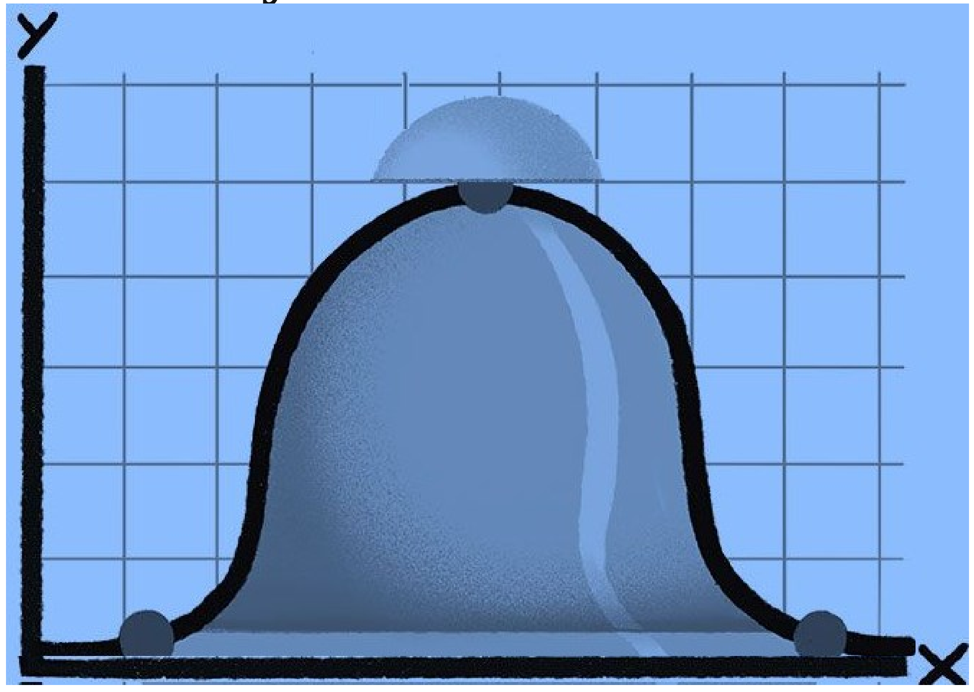
$$P_{(k=0)} = \frac{\lambda^k \cdot e^{-\lambda}}{k!} = \frac{3^0 \cdot e^{-3}}{0!} = 0.05$$

This low probability suggests that the proposed lifetime **10^{33} yrs** is **too short**.
This is not a realistic assessment of the probability of observations.

Normal Distribution (Gaussian Distribution)-Bell Curve

- Bell curve is a common type of distribution for a variable, also known as Normal distribution.
- In Normal Distribution, "Bell curve" originates from the fact that the graph depict a symmetrical bell-shaped curve.
- Highest point on the curve, or the top of the bell, represents the **most probable event** in a series of data (its Mean, Mode and Median are same in this case), while all other possible occurrences are symmetrically distributed around the mean, creating a downward-sloping curve on each side of the peak. The width of the bell curve is described by its **SD**.

- The term "bell curve" is used to describe a graphical depiction of a Normal probability distribution, whose underlying **SDs** from the Mean create the curved bell shape.
- **SD** is a measurement used to quantify the variability of data dispersion, in a set of given values around the Mean.
- Mean, in turn, refers to the average of all data points in the data set and will be found at highest point on the bell curve.



Normal Distribution

Example. Using normal distribution, explain the SAT scores obtained from the students. Where the data follows a normal distribution with a mean score (M) of 1150 and a standard deviation (SD) of 150.

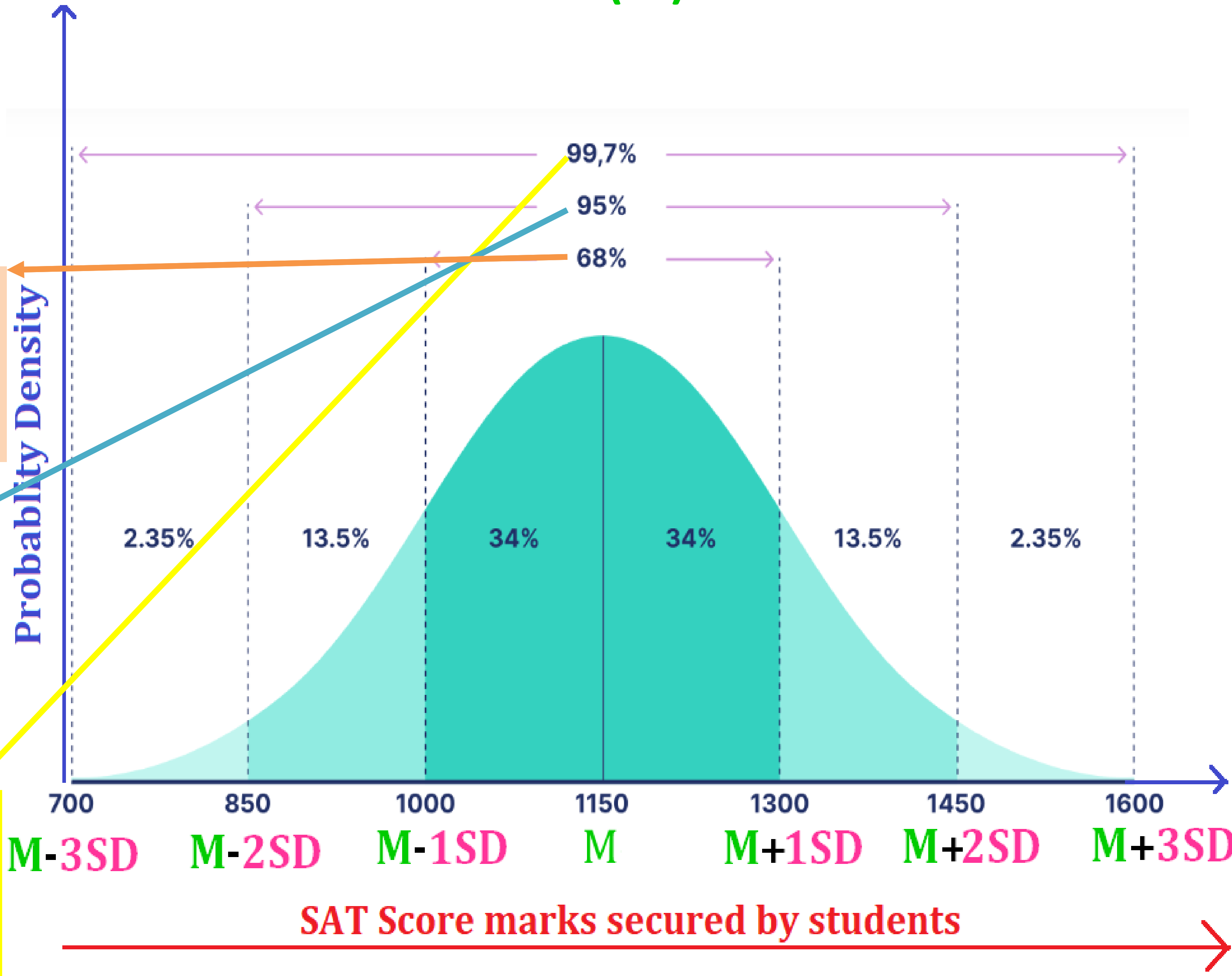
Solution: Collect the scores

Using Normal Distribution:

- Around **68%** of scores are between 1,000 & 1,300, with **1SD** above and below the **Mean**.

- Around **95%** of scores are between 850 & 1,450, with **2SD** above and below the **mean**.

- Around **99.7%** of scores are between 700 & 1,600, **3SD** above and below the **mean**. with



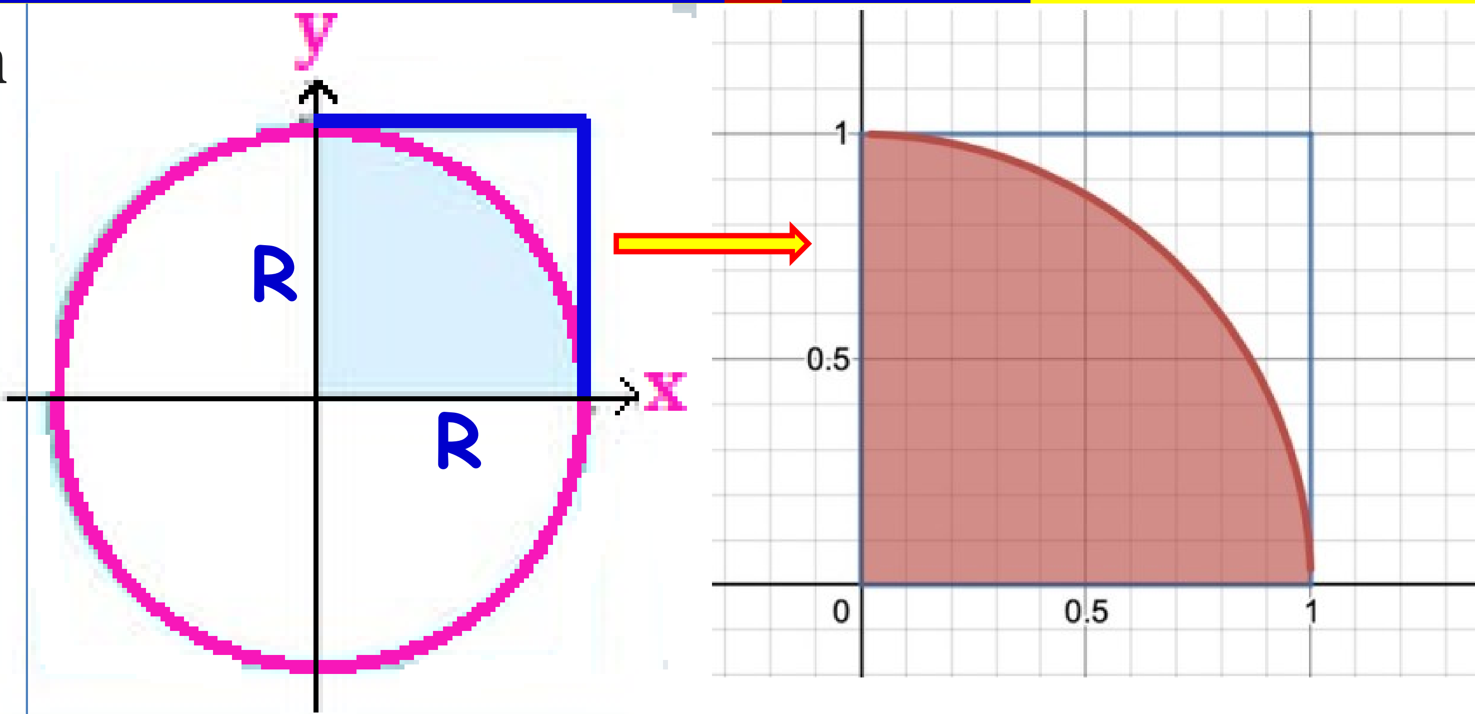
Monte-Carlo Method: Determination of "π" Value

Monte Carlo method tends to follow a particular pattern:

1. **Define** a domain of possible inputs
2. **Generate** inputs randomly over the domain
3. **Perform** a deterministic computation on the inputs
4. **Aggregate** the results

Determination of "π" Value:

For example: consider a quadrant (circular sector) inscribed in a unit square. Given that the ratio of their areas is $\pi/4$, the value of π can be approximated using a Monte Carlo method:



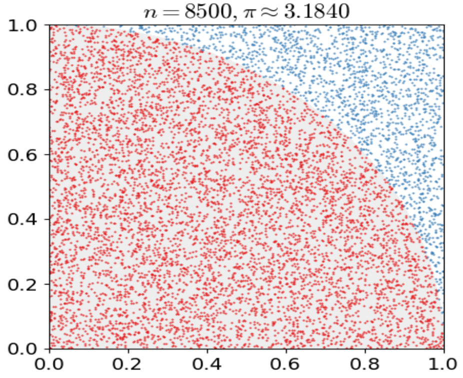
1. **Draw** a square, then Inscribe a quadrant within it
2. Uniformly **scatter** a given number of points over the square
3. **Count** the number of points inside the quadrant, i.e. having a distance < 1
4. The ratio of inside-count & total-sample-count is an estimation of the two areas.

Multiply the result by 4 to estimate "π" Value

Example. $= \dfrac{n\,(Quadr\ Area)}{n\,(Square\ Area)} = \dfrac{10}{13} = 0.769$

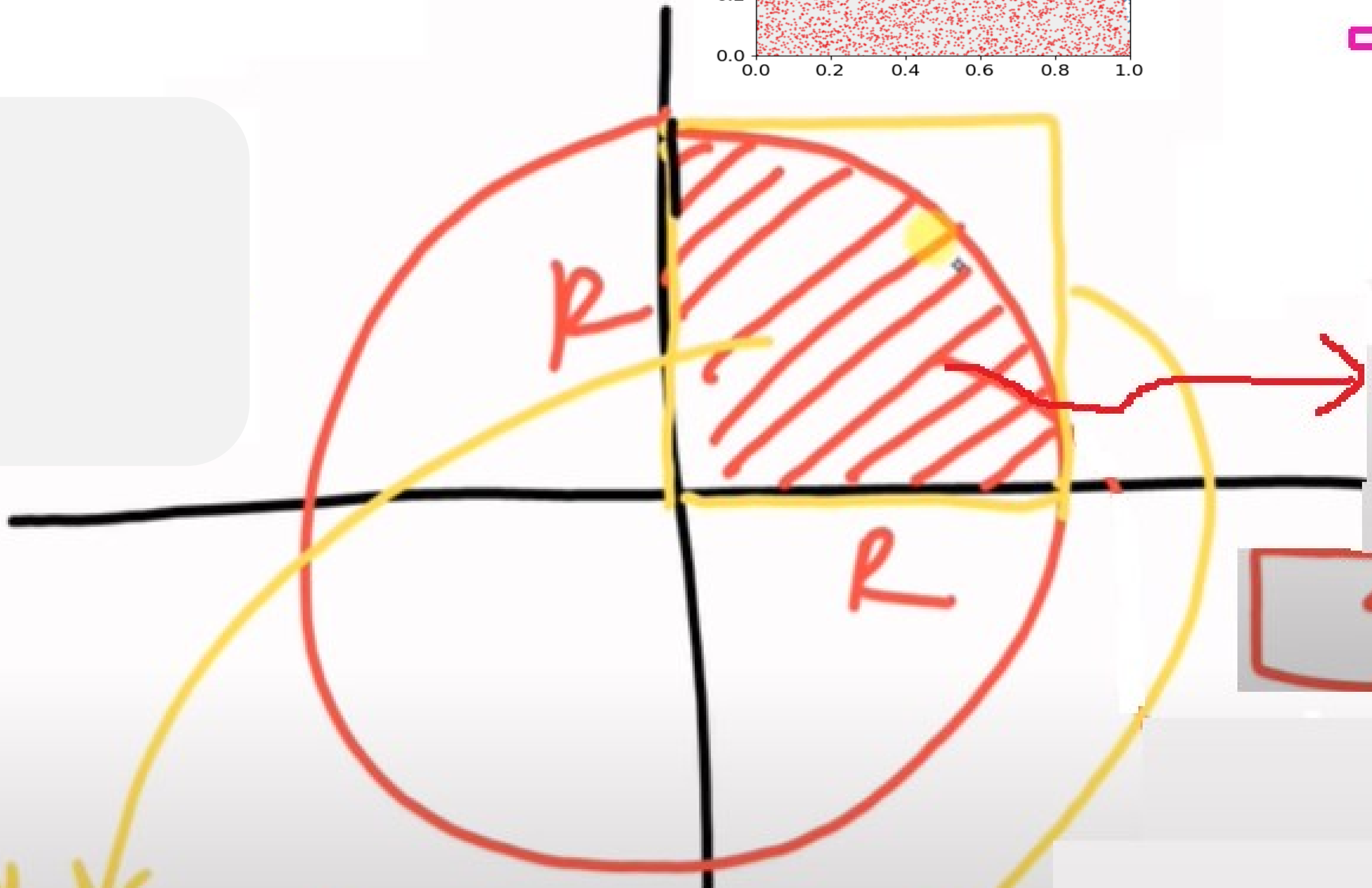
$$\therefore \boldsymbol{\pi} = 4 \times \frac{n\,(Quadr\ Area)}{n\,(Square\ Area)} = 4 \times 0.769$$

$$\boldsymbol{\pi} = 3.077$$



$$A = \pi \cdot R^2$$

$$\Rightarrow \pi = \frac{A}{R^2}$$



$$Q = \frac{1}{4} A$$

$$\boxed{4Q = A}$$

$$\pi = \frac{4Q}{R^2}$$

(1)

$$\frac{Q}{R^2} = \frac{\text{area of } \frac{1}{4} \text{ circle}}{\text{area of square } R} \cong \frac{n\,(Quadr\ Area)}{n\,(Square\ Area)}$$

Numerical Problems

(**Poisson Distribution** : Calculating the number of occurrences of an event in a given duration)

Example 1. The number of particles emitted per second by a random radioactive source has a Poisson distribution with $\lambda = 4$. Calculate the probability of $P(X = 0)$, $P(X = 1)$ and $P(X = 2)$.

Given: $\lambda = 4$

Solution: We know that, $P_{(X=k)} = \dfrac{\boldsymbol{\lambda}^k \cdot e^{-\boldsymbol{\lambda}}}{k!}$

(a) $P(X = 0)$, $P_{(k=0)} = \dfrac{4^0 \cdot e^{-4}}{0!} = \dfrac{1 \cdot e^{-4}}{1} = 0.018$

(b) $P(X = 1)$, $P_{(k=1)} = \dfrac{4^1 \cdot e^{-4}}{1!} = \dfrac{4 \cdot e^{-4}}{1} = 0.073$

(c) $P(X = 2)$, $P_{(k=2)} = \dfrac{4^2 \cdot e^{-4}}{2!} = \dfrac{16 \cdot e^{-4}}{2} = 0.144$

Example 2. On a particular place, volcanic eruption occurs once every 100 years on average. Calculate the probability of k = 0, 1, and 2 volcanic eruption in a 100 year interval, assuming the Poisson model is appropriate.

Given: $\lambda = 1$

Solution: We know that,

$$P_{(X=k)} = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

(a) $P(k = 0)$,

$$P_{(k=0)} = \frac{1^0 \cdot e^{-1}}{0!} = \frac{1 \cdot e^{-1}}{1} = 0.368$$

(b) $P(k = 1)$,

$$P_{(k=1)} = \frac{1^1 \cdot e^{-1}}{1!} = \frac{1 \cdot e^{-1}}{1} = 0.368$$

(c) $P(k = 2)$,

$$P_{(k=2)} = \frac{1^2 \cdot e^{-1}}{2!} = \frac{1 \cdot e^{-1}}{2} = 0.184$$


Thank you for being good students. I know you are headed for greater heights. Best of luck as you move to the next class.